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|---|---|---|
| | **GLOBAL ECONOMICS**<br>**GRADE: 10**<br>**LEARNING EVIDENCE 2**<br>**DECISION-MAKING UNDER UNCERTAINTY**<br>**TEACHER'S NAME: Nicolás López Cuéllar** | **1ST TERM**<br>**2026–2027** |

| | |
|---|---|
| **Learning objective:** Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative. | **Assessment criteria:**<br>C5: Constructs opportunity loss tables and applies the Minimax regret criterion to select the alternative that minimises the maximum opportunity loss.<br>C6: Calculates Expected Monetary Value using stated or estimated probabilities and applies the Bayes decision criterion to select the alternative with the highest expected payoff.<br>C7: Uses subjective probabilities and sensitivity analysis to examine how changes in probability assumptions affect the preferred decision. |

## 1. Criteria assessment

Each problem is worth 2 points. Show all calculations, including the complete regret table for C5, every EMV for C6, and the threshold equation and decision ranges for C7. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. A minimum of 4 points is required to pass each criterion.

| Question | C5 | C6 | C7 |
|---|---|---|---|
| 1 | 2 | – | – |
| 2 | 2 | – | – |
| 3 | 2 | – | – |
| 4 | – | 2 | – |
| 5 | – | 2 | – |
| 6 | – | 2 | – |
| 7 | – | – | 2 |
| 8 | – | – | 2 |
| 9 | – | – | 2 |
| **Total** | **6** | **6** | **6** |

## 2. 1. Community Garden Greenhouse [C5-2]

A community garden must choose one of three greenhouse plans. Annual net benefits are in thousands of dollars.

| Alternative | Excellent | Average | Poor |
|---|---|---|---|
| Large ($A_1$) | 70 | 30 | -10 |
| Medium ($A_2$) | 55 | 45 | 20 |
| Small ($A_3$) | 35 | 40 | 32 |

What decision would be selected using the Minimax Regret criterion? Justify your answer.

## 3. 2. Coastal Ferry Capacity [C5-2]

A ferry operator is choosing a capacity plan. Seasonal profit is in thousands of dollars; no reliable probabilities are available.

| Alternative | High | Normal | Low |
|---|---|---|---|
| Add two ferries ($A_1$) | 100 | 35 | -25 |
| Add one ferry ($A_2$) | 75 | 55 | 10 |
| Lease peak capacity ($A_3$) | 55 | 48 | 30 |
| Current fleet ($A_4$) | 30 | 30 | 30 |

What decision would be selected using the Minimax Regret criterion? Justify your answer.

### 4. 3. Mobile Clinic Schedule [C5-2]

A health organization must select a mobile-clinic schedule. Net community value is in thousands of dollars.

| Alternative | Very high | High | Moderate | Low |
|---|---|---|---|---|
| Full ($A_1$) | 90 | 50 | 20 | -20 |
| Flexible ($A_2$) | 70 | 60 | 45 | 10 |
| Limited ($A_3$) | 45 | 48 | 42 | 35 |

What decision would be selected using the Minimax Regret criterion? Justify your answer.

### 5. 4. Urban Farm Crop Mix [C6-2]

An urban farm is selecting a crop mix. Profits are in thousands of dollars.

| | Strong | Stable | Weak |
|---|---|---|---|
| $P(S_j)$ | 0.50 | 0.30 | 0.20 |
| Specialty ($A_1$) | 80 | 40 | -10 |
| Balanced ($A_2$) | 60 | 55 | 20 |
| Staple ($A_3$) | 40 | 45 | 35 |

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

### 6. 5. Museum Exhibition Scale [C6-2]

A museum is deciding the scale of a temporary exhibition. Payoffs are in thousands of dollars.

| | High | Expected | Low |
|---|---|---|---|
| $P(S_j)$ | 0.25 | 0.45 | 0.30 |
| Major ($A_1$) | 95 | 35 | -20 |
| Standard ($A_2$) | 70 | 50 | 15 |
| Compact ($A_3$) | 50 | 45 | 32 |
| Collection only ($A_4$) | 25 | 25 | 25 |

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

### 7. 6. Solar Installer Staffing [C6-2]

A solar installer is choosing a staffing plan. Payoffs are in thousands of dollars.

| | Surge | Strong | Steady | Slow |
|---|---|---|---|---|
| $P(S_j)$ | 0.15 | 0.35 | 0.30 | 0.20 |
| Rapid expansion ($A_1$) | 100 | 60 | 20 | -30 |
| Moderate expansion ($A_2$) | 75 | 65 | 45 | 10 |
| Current staff ($A_3$) | 50 | 52 | 48 | 40 |

What decision would be selected using the EMV or Bayes criterion? Justify your answer.

### 8. 7. Festival Catering Contract [C7-2]

A caterer must choose between a large and a small contract. Payoffs are in thousands of dollars.

| Alternative | High ($S_1$) | Low ($S_2$) |
|---|---|---|
| Large ($A_1$) | 80 | 10 |
| Small ($A_2$) | 45 | 35 |

Let $P(S_1) = p$. State the preferred alternative for every probability value ($0 \le p \le 1$).

### 9. 8. Electric-Bike Launch [C7-2]

A retailer is comparing an aggressive launch with a cautious launch. Payoffs are in thousands of dollars.

| Alternative | High | Moderate | Low |
|---|---|---|---|
| Aggressive ($A_1$) | 90 | 30 | -10 |
| Cautious ($A_2$) | 45 | 50 | 25 |

Suppose $P(S_1) = p$, $P(S_2) = \frac{p}{2}$. State the preferred alternative for every feasible probability value of $p$ and identify the threshold probability at which the retailer is indifferent between the two alternatives.

### 10. 9. Rainwater-System Bid [C7-2]

An engineering firm must choose between a standard bid and an innovative bid. Payoffs are in thousands of dollars.

| Alternative | Immediate | Revision | Delayed |
|---|---|---|---|
| Standard ($A_1$) | 20 | 55 | 75 |
| Innovative ($A_2$) | 70 | 40 | 30 |

Suppose $P(S_1) = 2p$, $P(S_2) = p$. State the preferred alternative for every feasible probability value of $p$ and identify the threshold probability at which the firm is indifferent between the two alternatives.